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street or public area; however, actual lighting needs may vary during the year because of the following reasons:

(i) Seasonal variations of daylight/night time hours; this depends on the geographical location of the area.

(ii) Changing weather conditions which influence perceived visual performance (i.e. dry or wet road surface).

(iii) Changing traffic density on the street or public area during the night (i.e. different temporal pattern of usage such as increased usage at 'rush hour') or following the fluctuation in social activity (i.e. school terms, national holiday periods).

(iv) Changing functionality of the street or public area (i.e. roads are closed for a certain period or turned to pedestrian areas in festive seasons).

Annual energy consumption indicator (AECI) shall be calculated with the following formula:

$$D_E = \frac{\sum_{j=1}^{m}\left(P_j \cdot t_j\right)}{A} \qquad \text{F11}$$

where

D_E = annual energy consumption indicator for a road lighting installation, in watt hours per square metre ($Wh.m^{-2}$)

P_j = operational power associated with the jth period of operation, in watts (W)

t_j = duration of jth period of operation profile when the power P_j is consumed, over a year, in hours (h)

A = size of the area lit by the same lighting arrangement, in m^2

m = number of periods with different operational power P_j. m shall also consider the period over which the non-active power is consumed. This period would generally be the time when the lighting is not operational, i.e. daylight hours and any night time period when the lighting is not lit

If the light output of a light source is intended to be constant, but the power consumption of this light source (or other electrical devices) varies in time [e.g. if constant light output (CLO) drivers are used], the average power consumption over the anticipated lifetime shall be included in the calculation. The calculation shall clearly indicate the lifetime assumptions used for calculation of the average consumption and how this value was evaluated.

Annual energy consumption indicator (D_E) shall be always presented and used together with the values of the power density indicator (D_P) for assessment of the energy performance of a particular lighting system.

F.3.2 Annual energy consumption indicator (AECI)—Calculation worked examples

F.3.2.1 Worked example without constant light output (CLO)

The following are worked examples of AECI calculations to show how the method is used in practice.

Example lighting scheme from Paragraph F2.5 above, with added lighting scheme operational parameters (numerical values are indicative only):

(a) Light point controller power = 2 W, 24/7 applicability.

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